

Sri Lanka Institute of Information Technology

Faculty of Computing

Department of Software Engineering



SE3070 : Case Studies in Software Engineering

Year 3 Semester 2 – 2026

Assignment 01

Case Study 2: Smart Disaster Early-Warning and Emergency Coordination System

Group: Group_050

Student Registration Number	Student Name	Email Address
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Introduction

Disaster response in Sri Lanka currently depends on fragmented methods such as phone calls, radio broadcasts, WhatsApp groups, and paper reports, which leads to delayed warnings, duplicated effort, and relief supplies that miss the areas of greatest need. This design proposes a unified digital platform for the Disaster Management Centre (DMC) that issues timely, location-specific hazard alerts to citizens and coordinates shelters, rescue teams, and relief supplies across government agencies, the armed forces, NGOs, and volunteers. The platform is designed to remain usable during power and network outages and to stay simple for civilians and field responders working under stressful, time-critical conditions.

Use Case Allocation

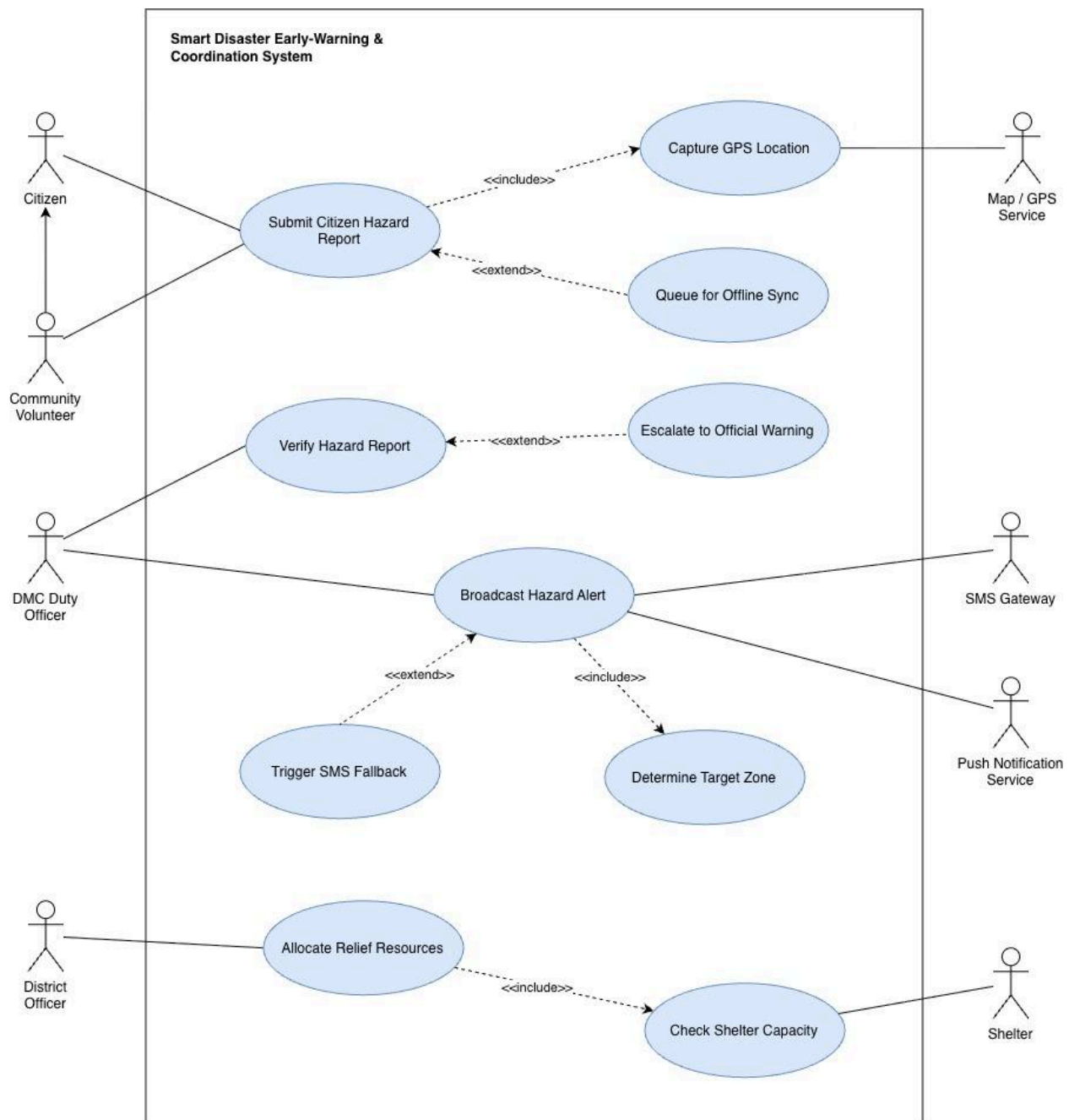
Each member's individual component (Section 2) is based on a different use case from the four identified in Section 1.1.

Student Registration Number	Student Name	Selected Use Case	Primary Actor
IT23733794	Senadheera K.M.N	Submit Hazard Report	Citizen / Field Volunteer
IT23713994	T.H.P Janith M	Broadcast Hazard Alert	DMC Officer
IT23714298	A.P.P Lanka	Verify Hazard Report	DMC Officer
IT23752368	Aathika I.F.	Allocate Relief Resources	District Officer

1. Group Component

This section was completed collaboratively by all group members.

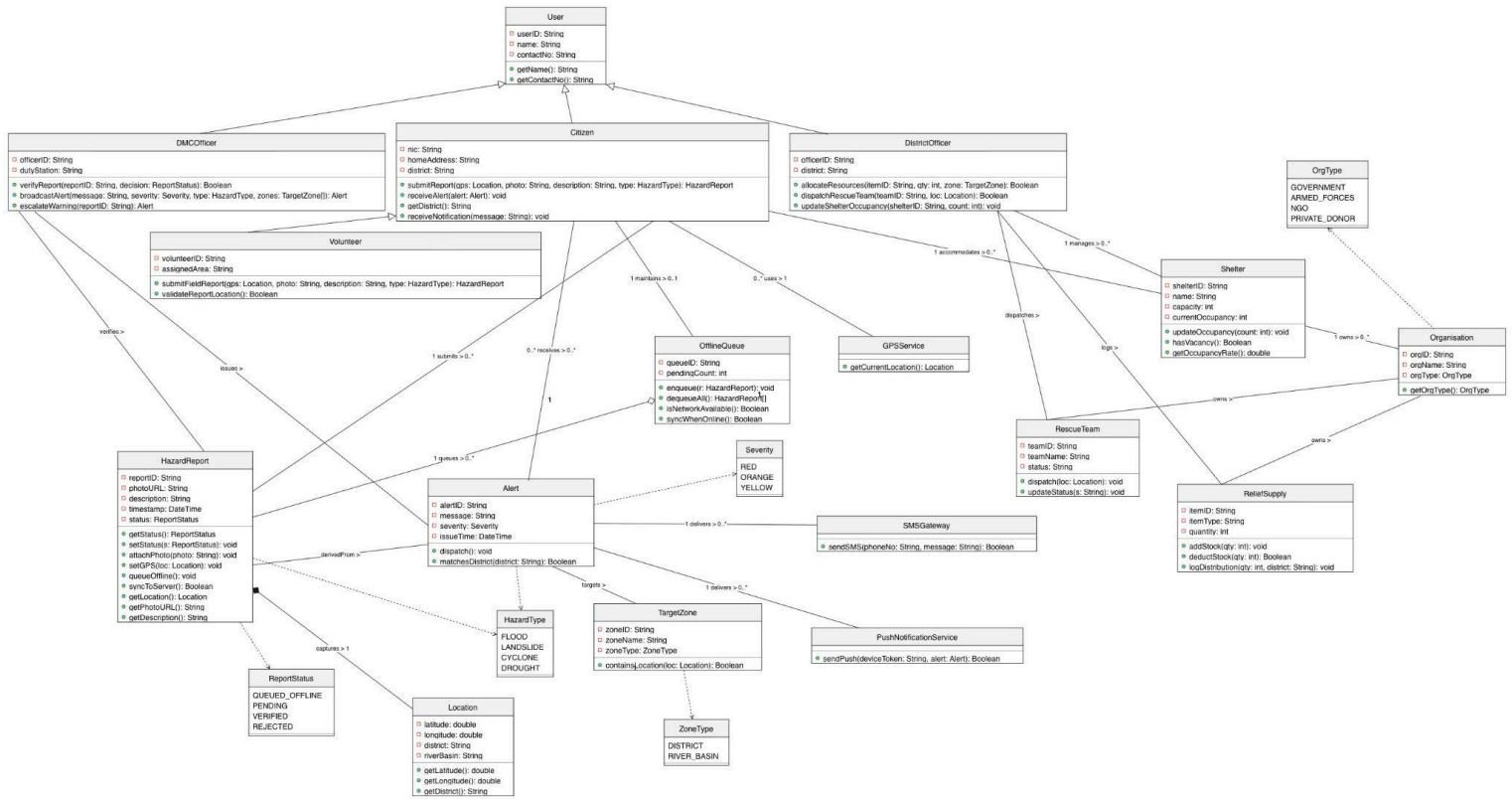
1.1 High-Level Use Case Diagram



View original image file [here](#)

1.2 Class Diagram

Must be fully consistent with the use case diagram above: every use case should map to at least one class method, with correct relationships and multiplicities shown.



View original image file [here](#)

2. Individual Components

Each member below produced their section independently, based on a different use case from Section 1.1. All sequence diagrams reference only the classes and methods defined in the Group Class Diagram (Section 1.2).

Senadheera K.M.N - Use Case: Submit Citizen Hazard Report

2.1.1 Detailed Use Case Scenario

Use Case: Citizen or field volunteer submits a hazard report from the ground

Use Case Scenario: Submit Citizen Hazard Report

Description:

This use case describes how a citizen or a community volunteer reports a hazard (flood, landslide, cyclone, or drought) they observe nearby. The reporter opens the app, the system captures their GPS location automatically, they attach a photo and a short description, and submit the report to the DMC. If the device has no network connection at the moment of submission, the report is saved to local storage and queued to sync automatically once connectivity returns, so a reporter working in a blackout zone is never blocked from filing a report.

Preconditions:

- The citizen or volunteer has the mobile app installed and has granted camera and location permissions.
- The reporter is physically at or near the hazard location.
- For volunteers, the app has the volunteer's `assignedArea` on file.

Main Flow:

- The citizen opens the app and taps "Report Hazard".
- The system calls `GPSService.getCurrentLocation()` to capture the reporter's GPS coordinates.
- The citizen takes or uploads a photo of the hazard.
- The citizen selects a hazard type (FLOOD, LANDSLIDE, CYCLONE, or DROUGHT) and types a short description.
- The citizen taps "Submit Report".
- The system calls `Citizen.submitReport(location, photo, description, hazardType)`, which creates a new `HazardReport`.
- The system timestamps the report and sets its status to PENDING.
- The system shows the citizen a confirmation screen and adds the report to the DMC officer's PENDING queue.

Alternate Flows:

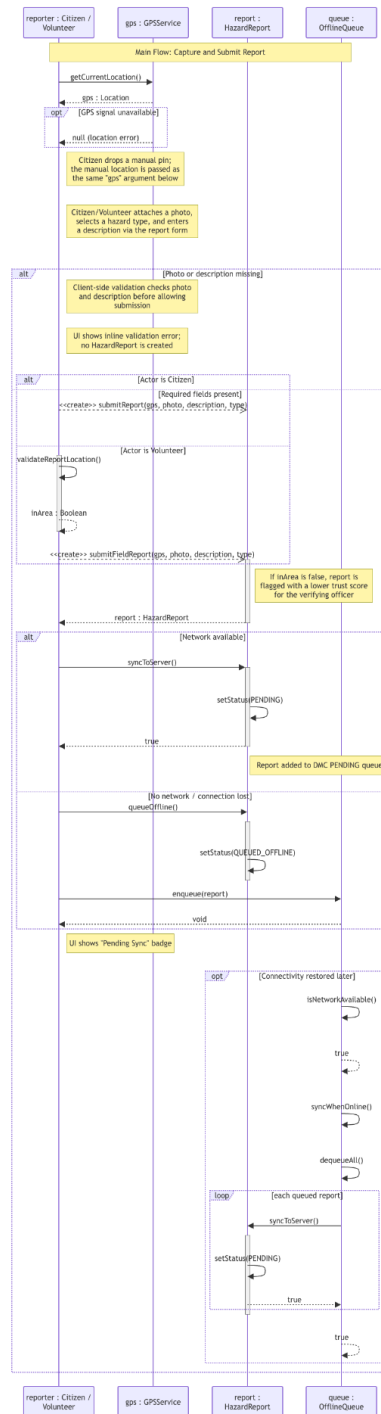
- GPS Signal Unavailable:
 - `GPSService.getCurrentLocation()` cannot get a fix (weak signal indoors or during a storm).
 - The system shows a "Location Not Found" message and lets the citizen drop a pin manually on the map instead.
 - The citizen confirms the manual pin and the flow continues from Main Flow step 3.
- Volunteer Submits Outside Assigned Area:
 - A community volunteer submits a report for a location outside their `assignedArea`.
 - The system still accepts the report but flags it with a lower trust score so the verifying officer reviews it more carefully.

Postconditions:

- A new `HazardReport` is created and saved with status PENDING (or QUEUED_OFFLINE if submitted without a network connection).
- The report's photo, GPS location, description, hazard type, and timestamp are all persisted.
- The reporter sees an on-screen confirmation that the report was received.

Exception Flows:

- No Network Connection (Offline Submission):
 - The device has no internet connection when the citizen taps "Submit Report".
 - The system detects the failed connection and calls `HazardReport.queueOffline()`, setting the report's status to QUEUED_OFFLINE.
 - The system calls `OfflineQueue.enqueue(report)` to store the report in local storage (SQLite) on the device.
 - The app shows a "Pending Sync" badge on the report instead of a submission error.
 - When the device regains connectivity, `OfflineQueue.syncWhenOnline()` calls `dequeueAll()` and `HazardReport.syncToServer()` for each queued report, updating status to PENDING once the server confirms receipt.
- Missing Photo or Description:
 - The citizen taps "Submit Report" without attaching a photo or entering a description.
 - The system shows a validation error and blocks submission until the required fields are filled.



2.1.2 Detailed Sequence Diagram

View original image file [here](#)

2.1.1 Interaction Design (UI/UX)

Storyboard



Low-Fidelity Wireframe

←

NEW REPORT

📞 117

👤

■ SYS.MOD: RPT-02 // FAST HAZARD

● OFFLINE ENGINE READY

STEP 1 OF 1 • RAPID INTAKE PROTOCOL

SCHEMA_V2.4

HAZARD DOCUMENTATION FORM

Verify automatic telemetry, bind visual evidence, and transmit to disaster logistics node.

01. OPTICAL VERIFICATION PAYLOAD

[OPTIONAL]

📷

[TAP TO CAPTURE PHOTO OR UPLOAD]

SUPPORTS RAW/JPEG/PNG • EXIF RETENTION ENABLED

RAW

evidence_094721.jpg

3.2MB • GPS EXIF BOUND

✕

02. SPATIAL COORDINATES

GNSS LOCK

📍 6.9271° N, 79.8612° E (±4m Accuracy)

Ward 07, Cinnamon Gardens, Colombo 07, Western Province

REF: LAT_6.9271 / LON_79.8612

📍

[(+) TARGET FIXED]

SCALE 1:5000

📍 [ADJUST PIN MANUALLY ON MAP]

03. HAZARD CLASSIFICATION MATRIX(MANDATORY)

☒

FLOOD

Flash / Overflow

☐

LANDSLIDE

Rockfall / Slope

☐

CYCLONE

Gale / High Winds

☐

DROUGHT

Water Deficit

04. SITUATION DESCRIPTORS & TELEMETRY

MULTI-SELECT

+ ROAD BLOCKED

+ RISING FAST

+ POWER DOWN

+ EVAC NEEDED

+ INJURIES NOTED

OBSERVATIONS & PRECISE DETAILS

e.g., Road submerged in 3 feet water, 4 families stranded near bridge...

ASCII_CHAR_LIMIT

0 / 250

☁️

SUBMIT HAZARD REPORT

📶 Auto-caches locally if no cellular connection is available.

[LOGO] DMC LANKA

● LIVE GPS

📞 117

👤

OK

✓

[LIVE SYNC COMPLETE]

Report Transmitted Successfully

Frontline situational data logged with National Emergency Network.

[STATUS: PENDING DMC REVIEW]

[REFERENCE IDENTIFIER]

NODE: LK-WEST-01

#DMC-LK-2025-9942

[COPY]

[RECEIPT: INCIDENT DATA LOG]

CHANNEL: SECURE-04

[CATEGORY]

Flash Flood (Kelani River Basin)

[COORDINATES]

Kelaniya, Gampaha District

6.9538° N, 79.9192° E (GPS Verified ±4m)

[TIMESTAMP]

Today, 14:32 PM

SYNC: INSTANT CLOUD

[DISPATCH ROUTE]

Western Province DOC & Sri Lanka Navy Rescue Unit

PAYLOAD: SHA-256 ENCRYPTED

STATUS: 200 OK

What happens next?

[PROTOCOL 3-STAGE]

1

Triage & Visual Verification

DMC technical team reviews photo telemetry & reported water gauge readings within 3–5 mins.

2

Field Responder Dispatch

Area Grama Niladhari, local police command, and assigned divisional response units are notified automatically.

3

Live Citizen Tracking

Receive push alerts and stage updates directly in your authenticated 'My Reports' timeline.

[ATTACHMENT VALIDATION]

1 FILE ATTACHED

[WIREFRAME PHOTO: KELANI-CULVERT-01.JPG]

EXIF GPS tagged • 1.4 MB • Status verified

📄 [View Report in My Reports]

🏠 [Return to Home Screen]

HAZARDS

RADAR

REPORT

SAFETY

High-Fidelity Wireframe

DMC LANKA
QUICK REPORT

ONLINE

117

New Hazard Report

Step 1 of 1 • Fast Submission

OFFLINE READY

Incident Photo Evidence

Optional • High Priority

Tap to Capture Photo
or browse device gallery

Clear ground photos assist rapid verification
by military and civilian rescue units.

IMG_2024_MONSOON.jpg

GPS Meta Extracted • Ready

Auto-Captured Coordinates

±4m (High Precision)

LIVE SPREAD LOOK

6.9271° N, 79.8612° E • Colombo 07, Western Prov

Adjust Pin Manually on Map (Rural / Faulty GPS)

Hazard Category

Select 1

Flood

Rising Water • Flash Floods

Landslide

Slope Failure • Earth Slips

Cyclone

High Winds • Gale Surges

Drought

Water Scarcity • Crop Loss

Hazard Situation Details

0 / 250

+ Road Blocked

+ Rising Fast

+ Power Lines Down

+ Evacuation Needed

e.g., Road submerged in 3 feet water, 4 families stranded near bridge. Need evacuation boat.

SUBMIT HAZARD REPORT

If offline, report is saved securely on this handset and transmitted automatically to the Emergency Operations Center when cellular signal returns.

Hazards

Radar

Report

Safety

DMC LANKA
HAZARD FEED

ONLINE

117

LIVE SYNC

Report Transmitted Successfully

Your frontline situational data is logged with the National Emergency Network.

STATUS: PENDING DMC REVIEW

REPORT REFERENCE

#DMC-LK-2025-9942

Copy

INCIDENT LOG RECEIPT

Verified

Incident Category

Flash Flood (Kelani River Valley)

Location & Coordinates

Kelaniya, Gampaha District

GPS: 6.9538° N, 79.9192° E

Timestamp

Today, 14:32 PM

Sync Type

Instant Cloud

Operational Dispatch Route

Forwarded to Western Province Disaster Operations Centre & Sri Lanka Navy Rescue Unit

What happens next?

DMC operations team evaluates photo and water level assessment within 3–5 minutes.

Field responder or Grama Niladhari notified for immediate ground verification.

You will receive direct status updates and safety alerts in your 'My Reports' feed.

Life Safety Warning

If water is rising rapidly, dial 117 or 119 immediately.

View Report in My Reports

Return to Home Screen

Hazards

Radar

Report

Safety

A P P Lanka - Use Case: Verify Hazard Report

2.1.1 Detailed Use Case Scenario

Use Case: DMC officer reviews and verifies citizen hazard reports

Use Case Scenario: Verify Hazard Report

Description:

This use case describes how a DMC duty officer checks a hazard report submitted by a citizen. The officer looks at the photo and GPS location to decide if the report is real. If it is real, the officer marks it as "VERIFIED". If it is fake or a mistake, they mark it as "REJECTED".

Preconditions:

- The DMC officer is logged into the central dashboard.
- There is at least one new report with the status "PENDING" in the system.
- The report has a photo, GPS location, and description.

Main Flow:

- The system shows a list of "PENDING" reports to the officer.
- The officer clicks on a report to open it.
- The system shows the report details (photo, map location, description, and time).
- The officer checks the photo and map location to see if it is a real hazard.
- The officer decides the report is real and clicks the "VERIFIED" button.
- The system changes the report status from PENDING to VERIFIED.
- The system saves a log with the officer's ID and the time of verification.
- The system removes the report from the "PENDING" list and shows it on the main map.

Alternate Flows:

- Report is Fake or Mistake (Rejected):
 - The officer sees the report is not a real hazard (e.g., just a puddle, not a flood).
 - The officer clicks "REJECTED" and types a short reason.
 - The system changes the status to "REJECTED" and sends a notification to the citizen.
- Escalate to Official Warning:
 - The verified report shows a very dangerous situation (like a big flood or landslide).
 - The officer clicks "Escalate to Warning".
 - The system creates a draft "Hazard Alert" so the officer can broadcast it to citizens.

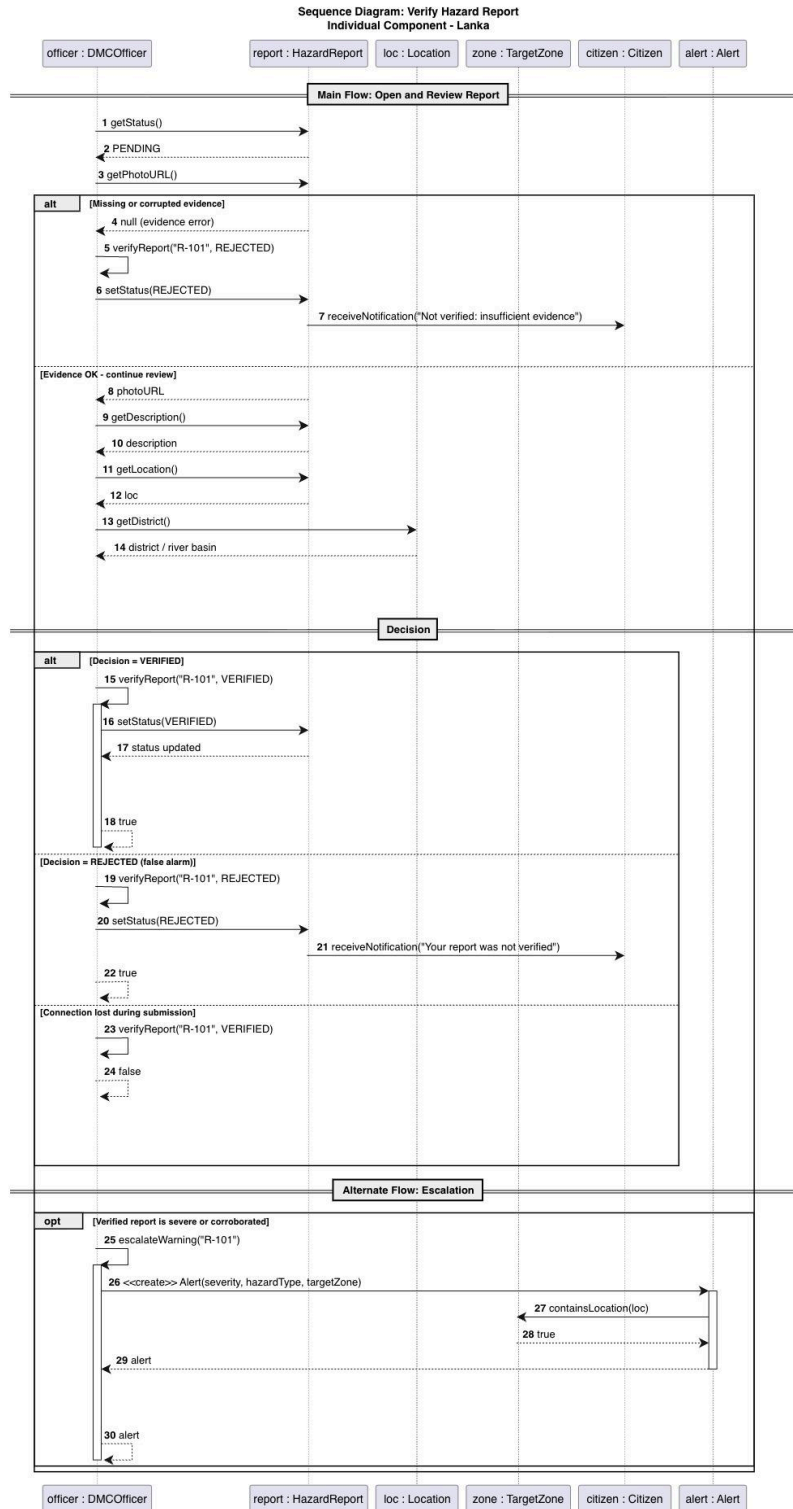
Postconditions:

- The report status is updated to "VERIFIED" or "REJECTED".
- The system saves a log of the officer's decision.
- The report is no longer in the "PENDING" list.

Exception Flows:

- Missing Photo or Bad GPS:
 - The system cannot load the photo, or the GPS location is missing/wrong.
 - The system shows an "Evidence Error" message.
 - The officer rejects the report because there is "not enough proof".
- Internet Connection Lost:
 - The dashboard loses connection to the server while the officer is clicking "Verify".
 - The system shows a "Connection Lost" error and does not save the decision.
 - The officer must wait and click "Verify" again when the internet is back.

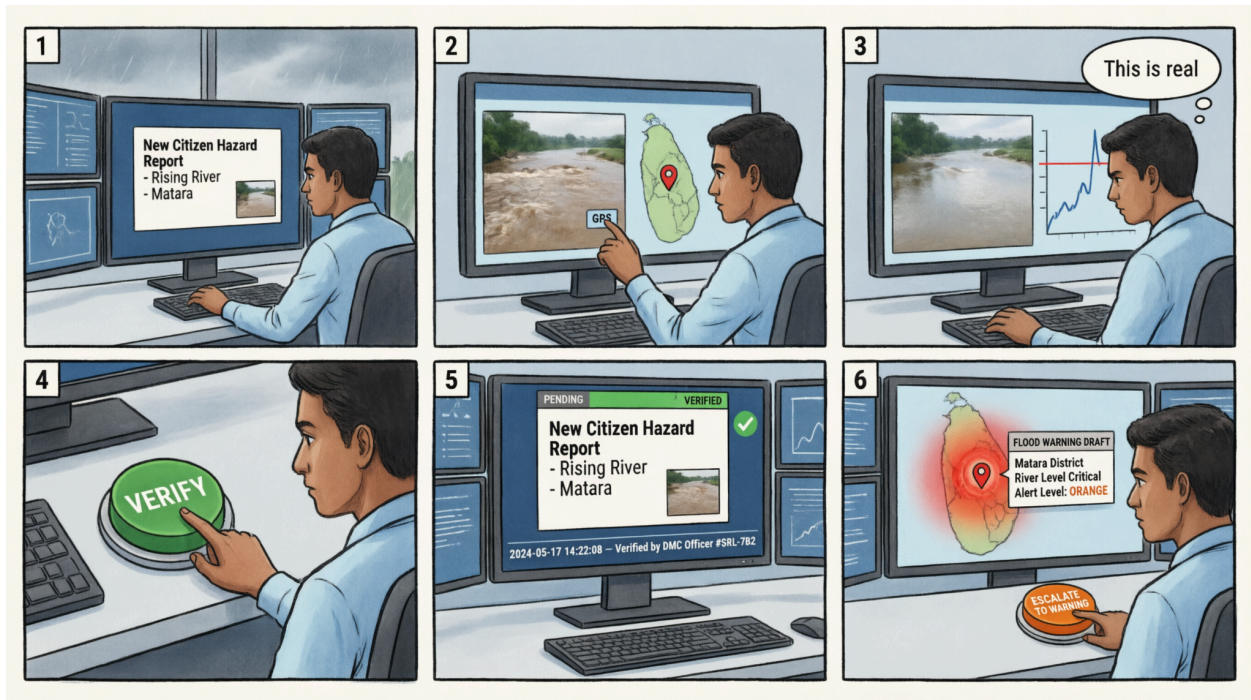
2.1.2 Detailed Sequence Diagram



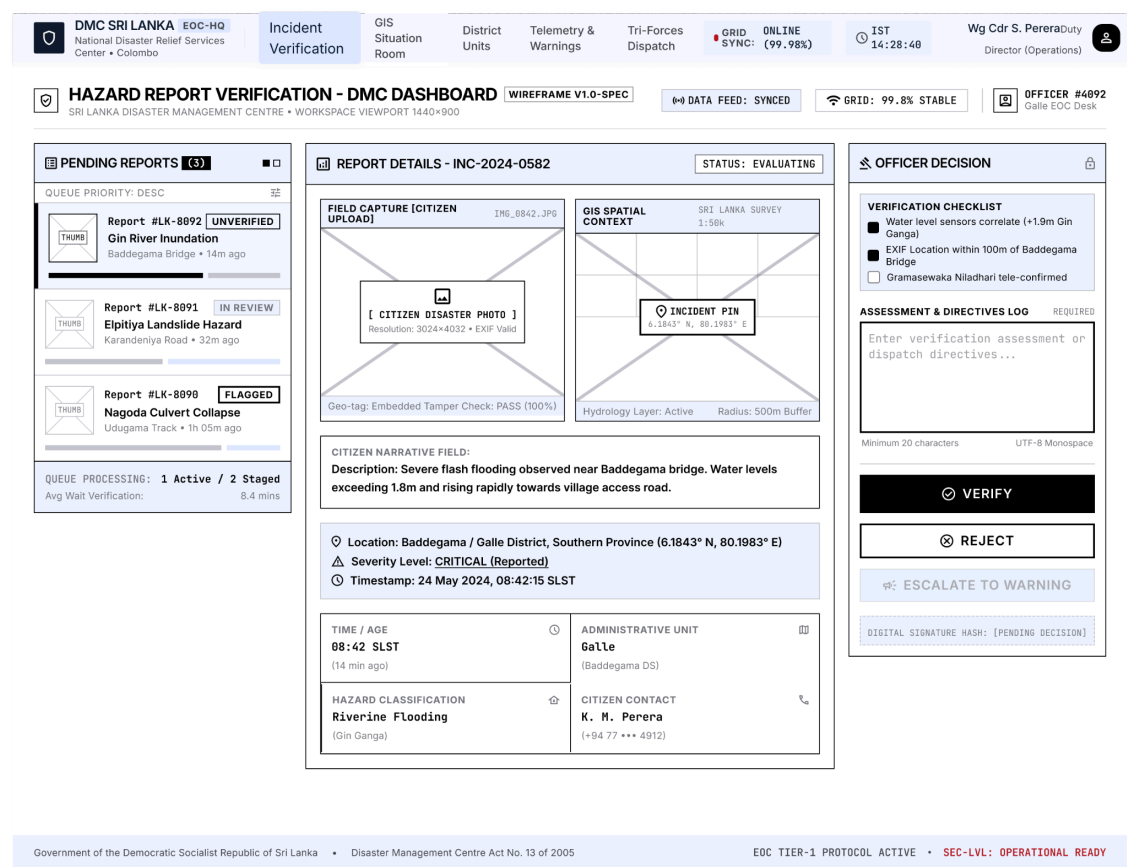
View original image file [here](#)

2.1.1 Interaction Design (UI/UX)

Storyboard



Low-Fidelity Wireframe



Aathika I.F. - Use Case: Allocate Relief Resources

2.1.1 Detailed Use Case Scenario

Use Case: District Officer allocates relief supplies and transport to affected shelters

Use Case Scenario: Allocate Relief Resources

Description:

This use case describes how a District Officer reviews shelters and hazard zones under their district, checks the relief stock available at the district warehouse, and allocates supplies (dry rations, water, tents, medical kits) and, where needed, a rescue/transport team to the location with the greatest need. The officer confirms the allocation, the system deducts warehouse stock and logs the distribution, and the shelter's occupancy record is updated.

Preconditions:

- The District Officer is logged into the district resource dashboard.
- At least one shelter or target zone has an open relief request with status "AWAITING ALLOCATION".
- The district warehouse stock ledger (ReliefSupply) is synced and reachable.

Main Flow:

- The system shows the list of shelters/zones needing supply, ranked by occupancy and severity.
- The officer selects a shelter to review its request details.
- The system shows the shelter's occupancy rate, vacancy status, request note, and GIS location.
- The officer confirms the request location falls within a valid target zone.
- The officer selects the item type and quantity to allocate and enters allocation notes.
- The officer clicks "Allocate Resources".
- The system deducts the allocated quantity from the warehouse stock and logs the distribution against the district.
- The system updates the shelter's occupancy/allocation record and marks the request as "ALLOCATED".

Alternate Flows:

- Warehouse Stock Insufficient:
 - The system finds the requested quantity exceeds the available warehouse stock.
 - The officer checks the partner organisation type (NGO / Armed Forces / Private Donor) linked to the district.
 - The system creates a partner resupply request instead of completing the allocation, and notifies the officer that the request has been forwarded.
- Emergency Transport Dispatch:

- The target zone is marked with CRITICAL severity and the shelter needs supplies moved immediately.
- The officer clicks “Dispatch Rescue Team” in addition to the allocation.
- The system assigns an available rescue/transport team to carry the allocated supplies to the shelter location and updates the team status to “EN ROUTE”.

Postconditions:

- The warehouse stock quantity is reduced by the allocated amount.
- A distribution log entry is saved against the item, quantity and district.
- The shelter’s occupancy/allocation status is updated and removed from the “Awaiting Allocation” queue.

Exception Flows:

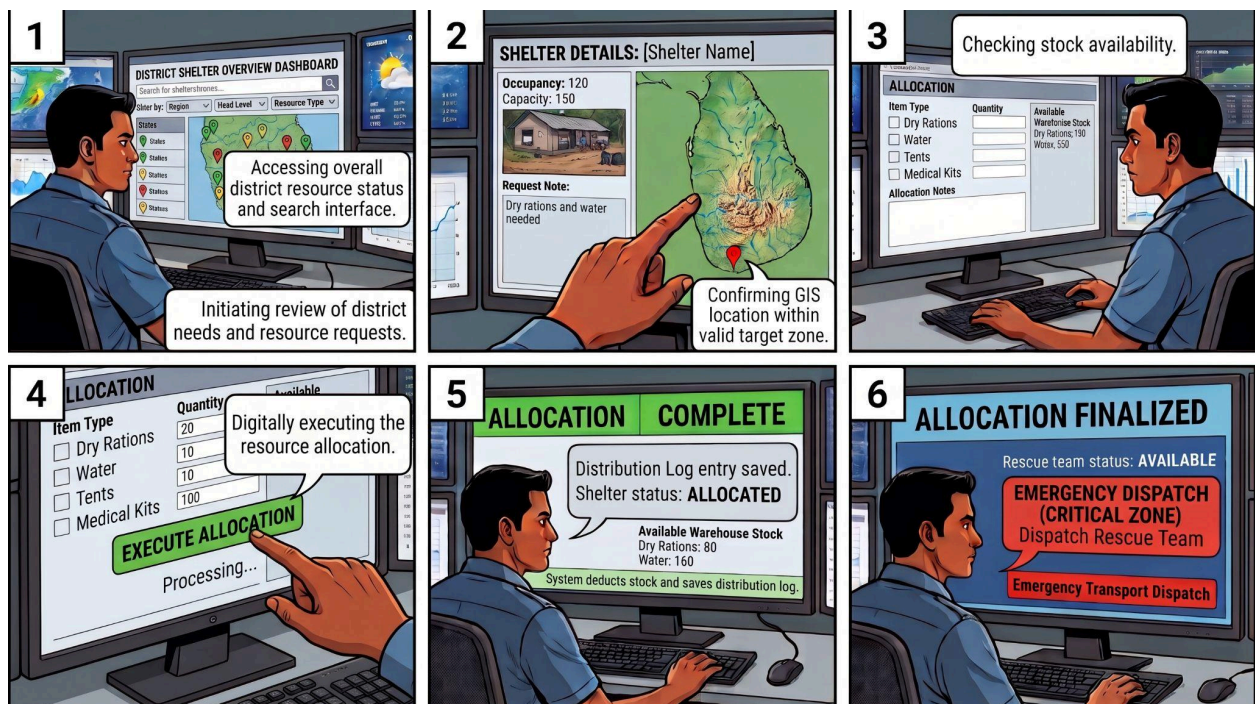
- Warehouse Ledger Sync Failure:
 - The system cannot reach the warehouse stock ledger to confirm current quantity.
 - The system shows a “Stock Sync Error” message and disables the “Allocate” button.
 - The officer must wait for the ledger to resync before allocating.
- Connection Lost During Allocation:
 - The dashboard loses connection to the server while the officer is clicking “Allocate Resources”.
 - The system shows a “Connection Lost” error and does not deduct stock or save the allocation.
 - The officer must confirm the stock level again and click “Allocate Resources” once the connection is restored.

2.1.2 Detailed Sequence Diagram

View original image file [here](#)

2.1.1 Interaction Design (UI/UX)

Storyboard



Low-Fidelity Wireframe

DMC

DMC SRI LANKA

DIST-OPS

Hazard Verification

Resource Allocation

GIS Situation Room

Shelter Registry

Tri-Forces Dispatch

GRID SYNC: (99.7%)

IST 09:14:02

Mrs. R. Aathika

District Officer

RELIEF RESOURCE ALLOCATION - DISTRICT DASHBOARD

GALLE DISTRICT SECRETARIAT - WORKSPACE VIEWPORT 1448x960

WIREFRAME V1.0-SPEC

(()) STOCK FEED: SYNCED

VEHICLES: 4 READY

OFFICER #00-1183

SHELTERS NEEDING SUPPLY

3

PRIORITY: OCCUPANCY DESC

Shelter #GL-014

Baddegama MV School - 6m ago

CRITICAL

Shelter #GL-009

Karadeniya Town Hall - 28m ago

HIGH

Shelter #GL-006

Udugama Temple Grounds - 1h 10m ago

MODERATE

QUEUE PROCESSING: 3 Active / 2 Staged

Avg. Dispatch Time: 22.6 mins

ALLOCATION DETAILS - GL-014

STATUS: EVALUATING

WAREHOUSE STOCK (DIST-04)

[STOCK LEVEL CHART]

Dry rations, water, tents

Last sync: 6 min ago - Source: RS-08

SHELTER LOCATION

[SHELTER PIN]

6.1560° N, 80.2010° E

Route: 12.4km - Est. transit 34 min

SHELTER REQUEST NOTE: Occupancy at 96% (192 / 200 pax). Dry ration stock at shelter exhausted since this morning; drinking water critically low. Requesting immediate resupply and one additional water bowser.

Location: Baddegama MV School, Baddegama DS, Galle District

Severity: CRITICAL (Reported)

Timestamp: 03 Sep 2026, 08:52:10 SLST

OCCUPANCY / CAPACITY

192 / 200 (96%)

ADMINISTRATIVE UNIT

Baddegama DS Division

ITEM REQUESTED

Dry Rations, Water, Tents

SHELTER CONTACT

Mr. D. Karunaratne (+94 77 *** 2201)

OFFICER ALLOCATION

ITEM & QUANTITY

Dry Ration Pack x 200

TARGET ZONE

Baddegama Zone (GL-204)

☒ Warehouse stock confirmed (DIST-04)

☒ Shelter location verified on GIS layer

☐ Transport vehicle assigned

ALLOCATION NOTES

Enter allocation justification or dispatch notes...

Minimum 20 characters - UTF-8 Monospace

ALLOCATE RESOURCES

DISPATCH RESCUE TEAM

REQUEST FROM PARTNER ORG

DISPATCH REF: HAMB: [PENDING ALLOCATION]

Government of the Democratic Socialist Republic of Sri Lanka - Disaster Management Centre Act No. 13 of 2005

DIST-OPS TIER-2 PROTOCOL ACTIVE - SEC-LVL: OPERATIONAL READY

High-Fidelity Wireframe

DMC

DMC SRI LANKA

DIST-OPS

Hazard Verification

Resource Allocation

GIS Situation Room

Shelter Registry

Tri-Forces Dispatch

GRID SYNC: 99.7%

IST 09:14:02

Mrs. R. Aathika

District Officer

DMC District Command - Relief Resource Allocation

LIVE CHANNEL: GALLE-DIST-OPS2

Active Session: R. Aathika (District Officer - Galle District)

DEFCON 3

SHELTERS NEEDING SUPPLY

3 OPEN

ALL (3)

CRITICAL (1)

RESUPPLYING

Baddegama MV School

Galle District - 6m ago

#GL-014 - SELECTED

CRITICAL

Karadeniya Town Hall

Galle District - 28m ago

#GL-009

HIGH

Udugama Temple Grounds

Galle District - 1h 10m ago

#GL-006

MODERATE

Auto-Triage Active - DMC Resource Engine v3.8.1 - Sync: OK

#GL-014 Baddegama MV School (Relief Resupply)

AWAITING ALLOCATION

Baddegama DS Division - Galle District, Southern Province

SHELTER PHOTO - VERIFIED

WAREHOUSE DIST-04 - 18% STOCK

Overcrowded Shelter Hall

08:41 SLST - EXIF GPS Match

Baddegama Relief Warehouse

12.4 km - Est. transit 34 min

SHELTER REQUEST NOTE

Occupancy at 96% (192 / 200 pax). Dry ration stock exhausted since this morning; drinking water critically low. Requesting immediate resupply and one additional water bowser before nightfall.

Report Timestamp

08:52 SLST (03 Sep 2026)

14 min elapsed - PRIORITY Q

OCCUPANCY / CAPACITY

192 / 200 (96%)

ADMINISTRATIVE UNIT

Baddegama DS Division

ITEMS REQUESTED

Dry Rations - Water - Tents

SHELTER CONTACT

D. Karunaratne (+94 77 319 2201)

OFFICER ALLOCATION

READY

ITEM TYPE

Dry Ration Pack (Family, 5kg)

QUANTITY TO ALLOCATE

- 200 units +

TARGET ZONE

Baddegama Zone (GL-204)

☒ Warehouse stock confirmed (DIST-04)

☒ Shelter location verified on GIS layer

☐ Transport vehicle assigned

ALLOCATION NOTES

Confirmed with Baddegama Divisional Secretariat. Dispatching ration pallets and 1x water bowser from DIST-04 warehouse.

118 / 500 characters

ALLOCATE RESOURCES

DISPATCH RESCUE TEAM

REQUEST FROM PARTNER ORG

National Relief Audit Trail

Auto-logged to DMC-NRMS - Audit Ref #GL-7742

Officer ID: SL-DO-1183 - IP: 10.142.9.61

DIST-04 WAREHOUSE STOCK

LOW

18%

4

DRY RATIONS REMAINING

VEHICLES AVAILABLE

Government of the Democratic Socialist Republic of Sri Lanka - Disaster Management Centre Act No. 13 of 2005

DIST-OPS TIER-2 PROTOCOL ACTIVE - SEC-LVL: OPERATIONAL READY

T H P Janith M - Use Case: Broadcast Hazard Alert

2.1.1 Detailed Use Case Scenario

Use Case: DMC officer broadcasts an official hazard alert to citizens

Use Case Scenario: Broadcast Hazard Alert

Description:

This use case describes how a DMC duty officer sends an official hazard alert to citizens. The officer writes the alert message, chooses the hazard type, severity level, and the area that is affected. The system then sends the alert to all citizens inside that area and shows it on the public map.

Preconditions:

- The DMC officer is logged into the central dashboard.
- The officer has "Broadcast" permission (not all users can send alerts).
- There is a VERIFIED hazard report, or a draft alert created from the "Escalate to Warning" option.
- The notification service is connected and working.

Main Flow:

- The officer clicks the "Broadcast Alert" button on the dashboard.
- The system opens the alert form, already filled with details from the verified report (hazard type and location).
- The officer selects the hazard type (flood, landslide, high wind, etc.) and the severity level (Advisory, Warning, or Evacuation).
- The officer draws or selects the affected area on the map and sets a radius.
- The officer types the alert message and the safety instructions for citizens.
- The system shows a preview of the alert and the number of citizens who will receive it.
- The officer checks the preview and clicks "Confirm & Send".
- The system asks the officer to confirm one more time, because the alert cannot be taken back.
- The system sends the alert as a push notification and SMS to all citizens inside the selected area.
- The system shows the alert on the public hazard map with a coloured zone.
- The system saves a log with the officer's ID, the alert content, the area, and the time.
- The system shows a "Broadcast Successful" message with the delivery count.

Alternate Flows:

Save as Draft:

- The officer is not ready to send the alert yet.
- The officer clicks "Save Draft" instead of "Confirm & Send".
- The system saves the alert with the status "DRAFT" so the officer can open and send it later.

Update an Active Alert:

- The situation changes (for example, the flood water rises).
- The officer opens the active alert and clicks "Update".
- The officer changes the severity level or the affected area and sends it again.
- The system sends an updated alert to the citizens and marks the old one as "SUPERSEDED".

Cancel an Alert:

- The hazard is finished or the alert was sent by mistake.
- The officer clicks "Cancel Alert" and types a short reason.
- The system sends an "All Clear" message to the same citizens and removes the zone from the public map.

Postconditions:

- The alert status is "ACTIVE" (or "DRAFT" if it was only saved).
- All citizens inside the affected area have received the notification.
- The alert is visible on the public hazard map.
- The system saves a log of the broadcast for auditing.

Exception Flows:

Notification Service Failed:

- The push notification or SMS gateway is down.
- The system shows a "Delivery Failed" message and does not mark the alert as sent.
- The system puts the alert in a retry queue and tries to send it again automatically.
- The officer is told to use the backup channel (radio or TV) if the retry also fails.

Invalid Alert Area or Empty Message:

- The officer forgets to draw the area, or leaves the message box empty.
- The system shows a validation error ("Please select an affected area" / "Message cannot be empty").
- The system does not send the alert until the officer fixes the problem.

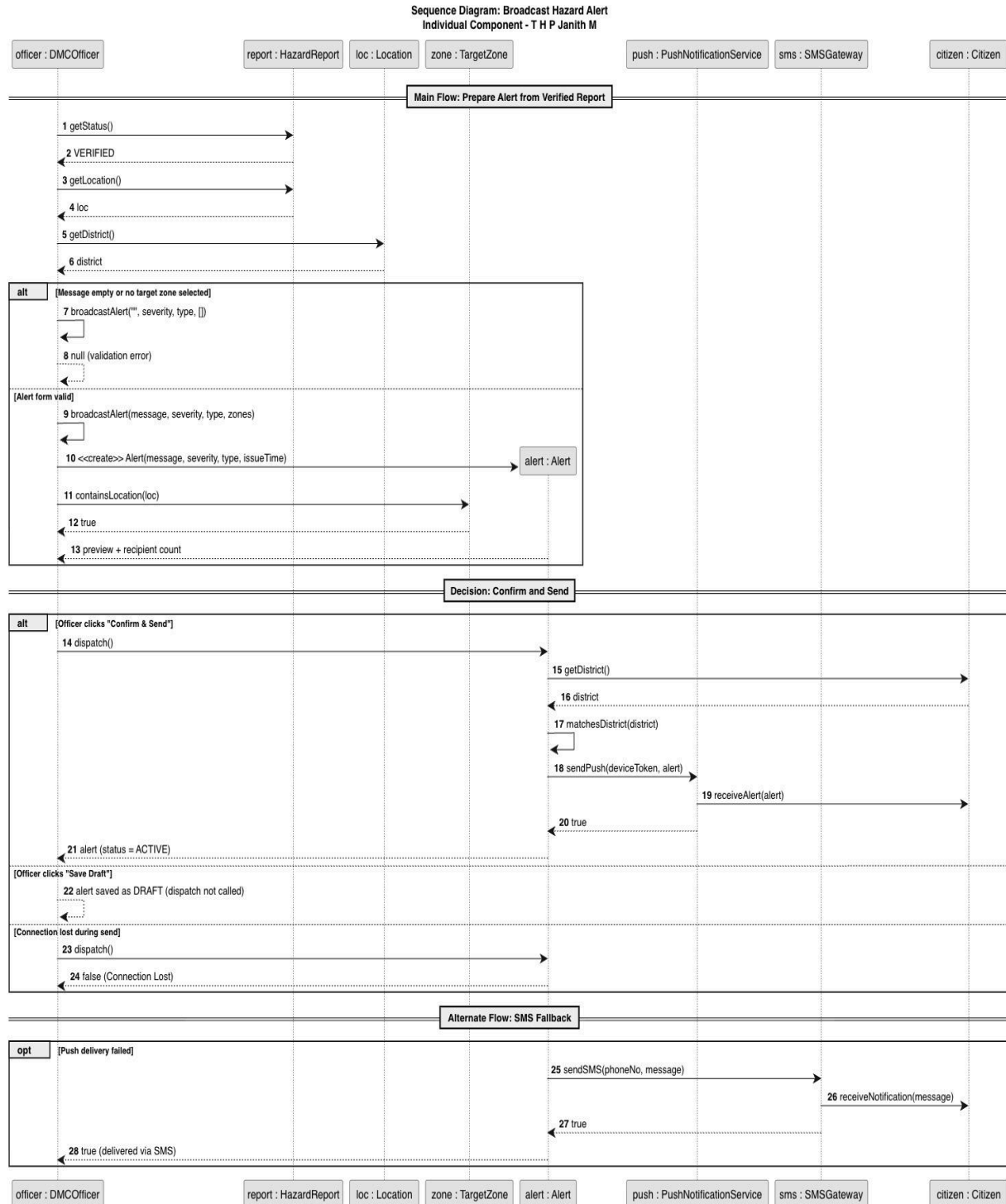
Internet Connection Lost:

- The dashboard loses connection to the server while the officer is clicking "Confirm & Send".
- The system shows a "Connection Lost" error and does not send the alert.
- The system keeps the alert as a draft so the officer does not lose the typed message.
- The officer sends it again when the internet is back.

Duplicate Alert:

- Another officer has already sent an alert for the same hazard and the same area.
- The system shows a "Similar Active Alert Exists" warning.
- The officer either cancels the new alert or chooses to update the existing one.

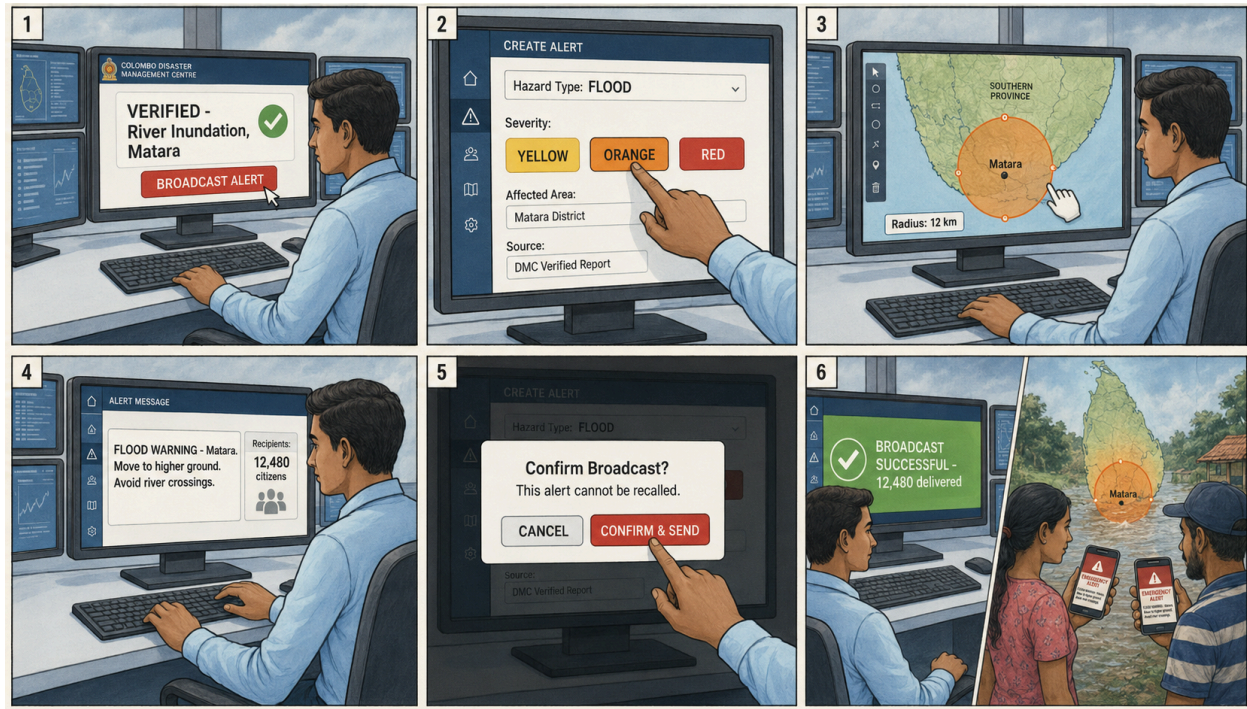
2.1.2 Detailed Sequence Diagram



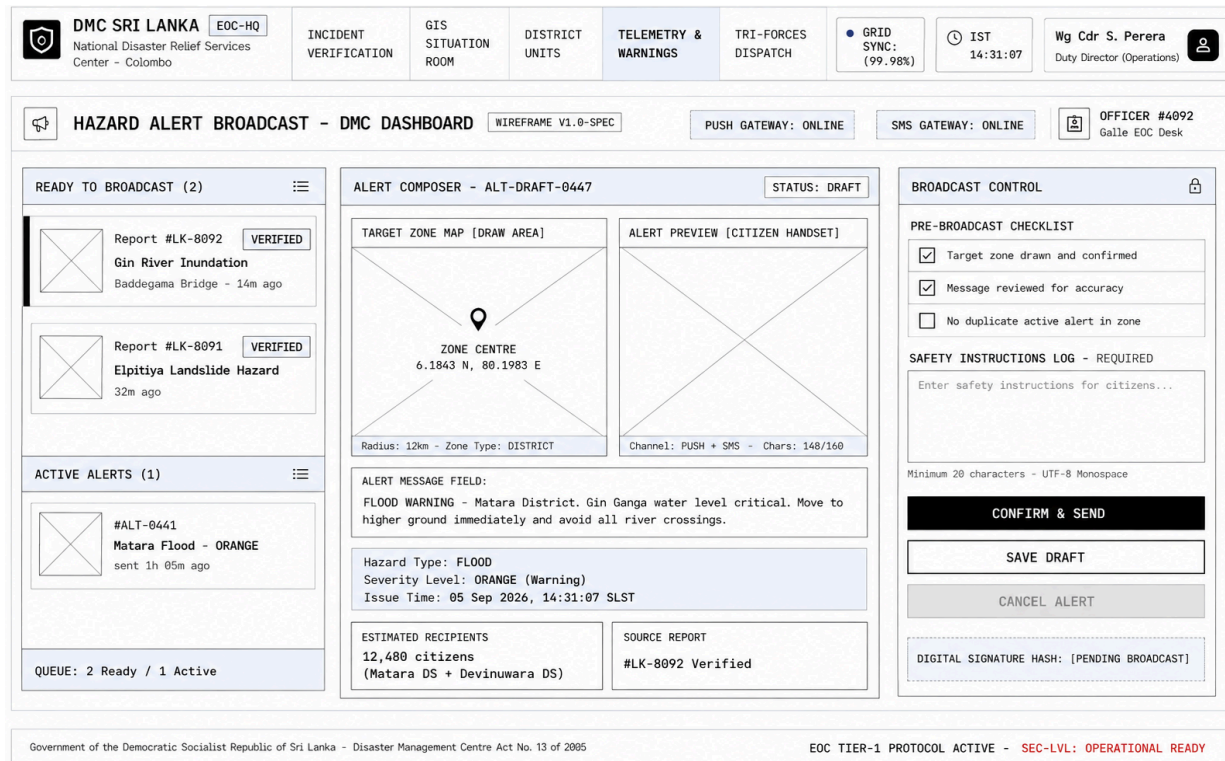
View original image file [here](#)

2.1.1 Interaction Design (UI/UX)

Storyboard



Low-Fidelity Wireframe



High-Fidelity Wireframe

DMC SRI LANKA

EOC-HQ

National Disaster Relief Services

Center - Colombo

INCIDENT VERIFICATION

GIS SITUATION ROOM

DISTRICT UNITS

TELEMETRY & WARNINGS

TRI-FORCES DISPATCH

GRID SYNC: ONLINE (99.98%)

IST 14:31:07

Wg Cdr S. Perera

Duty Director (Operations)

DMC Command Center - Hazard Alert Broadcast

LIVE CHANNEL: SOUTHERN-PROV-TAC4

Active Session: Major S. Jayawardena (Duty Officer - Southern Province)

DEFCON 2

READY TO BROADCAST

2 VERIFIED

ALL (3)

DRAFTS (1)

ACTIVE (1)

ORANGE

14m ago

Severe River Inundation

Galle District - Baddegama Basin

#LK-8092

SELECTED

RED

32m ago

Road Collapse - Galle

Galle Coastal - Route B412

#LK-8091

DRAFT

YELLOW

1h ago

Landslide Risk - Matara

Matara Tea Belt - Morawaka

#LK-8090

ACTIVE

AUTO-TRIAGE ACTIVE

DMC Engine v4.2.1

SYNC: OK

#ALT-DRAFT-0447

Flood Warning Broadcast (Gin Ganga)

Matara & Devinuwara DS Divisions, Southern Province

AWAITING CONFIRMATION

SOURCE REPORT

#LK-8092 - VERIFIED

TARGET ZONE SET - RADIUS 12 KM

12,480 recipients

LAT 06.1843 N - LON 80.1782 E

Southern Province Alert Layer

Zone Type: DISTRICT - Channels: PUSH + SMS

ALERT MESSAGE (CITIZEN FACING)

FLOOD WARNING - Matara District. Gin Ganga water level has passed the critical mark. Move to higher ground immediately, avoid all river crossings and stay away from riverbanks. Follow instructions from your Grama Niladhari.

ISSUE TIMESTAMP

14:31 SLST (05 Sep 2026)

drafted 6 min ago - PRIORITY Q

TARGET ZONES

Matara DS + Devinuwara DS

Southern Province, Zone ID GL-Z04

HAZARD CLASSIFICATION

Riverine Flooding

Gin Ganga Hydrographic Basin Code LK-S04

SOURCE OFFICER

K. M. Perera (Verifier)

Report #LK-8092 endorsed

BROADCAST CONTROL

READY

SELECTED SEVERITY LEVEL

ORANGE - WARNING

Immediate protective action advised

SAFETY INSTRUCTIONS

CHARS: 148/160

Coordinated with Matara DS. Rescue boats on standby at Gin Ganga station #4. CEB advised of transmission line risk.

CONFIRM & SEND

SAVE AS DRAFT

CANCEL ALERT

DELIVERY CHANNELS

☒ Push Notification - 9,140 devices

☒ SMS Fallback - 3,340 numbers

☐ Radio / TV backup relay

NATIONAL SECURITY AUDIT TRAIL

Auto-logged to DMC-NDMS - Audit Ref #ALT-0447

Officer ID: SL-DMC-8819 - IP: 10.142.8.24

GATEWAY STATUS

QUEUE HIGH

PUSH LATENCY

1.2 s (+0.4s)

SMS QUEUE

3,340 PENDING

Government of the Democratic Socialist Republic of Sri Lanka

Disaster Management Centre Act No. 13 of 2005

EOC TIER-1 PROTOCOL ACTIVE

SEC-LVL: OPERATIONAL READY